MILITARY SPECIFICATION

TRANSISTOR, PNP, GERMANIUM

TYPE 2N2834

## 1. SCOPE

1.1 Description.- This specification covers the detail requirements for a germanium, PNP transistor, and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Mechanical dimensions.- TO-3 (see figure 1).

1.3 Absolute maximum ratings.-

| $P_C$ $T_{Base}$ = 25°C | $T_J$ | $T_{stg}$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ |
|---|---|---|---|---|---|---|
| W | °C | °C | Vdc | Volts | Vdc | Adc |
| 85 (See 6.2) | 110 | -65 to +110 | 160 | 100 | 2 | 20 |

1.4 Primary electrical characteristics.-

| | $h_{FE}$ $I_C$ = 1.0 Adc | $h_{FE}$ $I_C$ = 10.0 Adc | $h_{fe}$ @ 5 mc | $V_{CE}$ (sat) $I_C$ = 20 Adc |
|---|---|---|---|---|
| Minimum | 50 | 25 | 2 | --- |
| Maximum | -- | 100 | --- | 0.5 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5960

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## 3. REQUIREMENTS

3.1 General.- Transistors shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Design, construction and physical dimensions.- Transistors shall be of the design, construction and physical dimensions shown on figure 1.

3.3 Performance characteristics.- Performance characteristics shall be as specified in 4.4, 4.5, and 4.6.

3.4 Marking.- Transistors shall be marked with the "USN" prefix in lieu of the "JAN" prefix and the type number shall be directly on the body. No color coding shall be allowed. Country of origin may be omitted.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.4, 4.5 and 4.6. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1).)

4.2 Inspection conditions.- Unless otherwise specified herein, all inspection shall be made at a base temperature of 25° ± 3°C.

4.3 Quality conformance inspection.- Quality conformance inspection shall consist of the examination and tests specified in 4.4, 4.5 and 4.6.

4.4 Group A inspection.- Group A inspection shall consist of the examinations and tests shown in table I. A device having one or more defects shall be counted as one defective.

4.5 Group B inspection.- Group B inspection shall consist of the examinations and tests shown in table II. Tests of each subgroup shall be performed in the order listed.

4.5.1 Salt atmosphere (corrosion).- The marking shall be legible after the test. There shall be no evidence (when examined with no magnification) of flaking or pitting of the finish or corrosion that will interfere with the mechanical or electrical application of the device.

4.5.2 Destructive tests.- Tests listed in subgroups 2, 3 and 4 are considered destructive.

4.5.3 Shock.- The shock testing apparatus shall be capable of providing shock pulses of the specified peak acceleration, waveform, and pulse duration to the body of the device. The acceleration pulse, as determined from the unfiltered output of a transducer with a natural frequency greater than 10,000 cycles per second, shall be a half sine waveform with an allowable distortion not greater than ± 20 percent of the specified peak acceleration. The pulse duration shall be measured between the points at 10 percent of the peak acceleration during rise and at 10 percent of the peak acceleration during decay. Absolute tolerance of the pulse duration shall be ± 40 percent of the specified duration.

4.6 Group C inspection.- Group C inspection shall consist of the tests shown in table III. Devices shall be subjected to the tests initially and at least once every 6 months, in lieu of the operating life test in Group B inspection shown in table II.

4.6.1 Safe Area test procedure.- The collector safe area shall be tested for three points in the circuit shown on figure 2 or equivalent, which fall on the boundaries of Safe Area:

Test 1.- The collector current shall be pulsed to 20 A at 50 V collector to emitter for 50 μsec. The frequency shall be 200 cps.

Test 2.- The collector current shall be pulsed to 30 mA at 150 V collector to emitter for 50 μsec. The frequency shall be 200 cps.

Test 3.- The collector current will be set at 0.15 A continuous (5 minutes or longer) at 50 volts collector to emitter with the device mounted on a heat sink capable of maintaining the case temperature at 60°C maximum.

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

6.2 Power-temperature derating versus safe area.- Figure 3 shows pulse safe areas. Figure 4 represents the allowable average power which could be dissipated at a given case temperature. The designer must consider both control curves. The collector to emitter voltage, collector current, and pulse width combination must fall within the pulse safe area control curve. In addition, the calculation of average power and case temperature must result in an operating point within the control curve shown on figure 4. In some instances, the allowable power at 25°C (case) is beyond the allowable safe areas. However, at higher case temperatures this would not be true. In any event, both average power-temperature derating and pulse safe areas must be observed. The proper control of temperature rise and peak power will insure that the maximum junction temperature is not exceeded. In addition, the proper control of pulse load line to stay within the applicable Safe Area curve will insure that a collector to emitter short will not be caused.

Preparing activity:
Navy - SH
(Project 5960-N520(NAVY))



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Table I - Group A inspection.

| Examination or Test | Conditions | | LTPD | Symbol | Limits | | Units |
|---|---|---|---|---|---|---|---|
| | MIL-STD-750 Method | Specific Conditions | | | Min | Max | |
| Subgroup 1 | | | 10 | | | | |
| Visual and mechanical examination | 2071 | | | | | | |
| Subgroup 2 | | | 5 | | | | |
| Collector to base cutoff current | 3036 Cond. D | $V_{CB}$ = -2V<br>$I_E$ = 0 | | $I_{CBO}$ | --- | -0.3 | mA |
| Collector to base cutoff current | 3036 Cond. D | $I_E$ = 0<br>$V_{CB}$ = -160V | | $I_{CBO}$ | --- | -10 | mA |
| Collector to emitter current | 3041 Cond. C | $V_{BE}$ = 0<br>$V_{CE}$ = -160V | | $I_{CES}$ | --- | -20 | mA |
| Collector to emitter cutoff current | 3041 Cond. A | $T_C$ = +85°C<br>$V_{BE}$ = 0.2V<br>$V_{CE}$ = -100 V<br>Pulsed 1/ | | $I_{CEX}$ | --- | -40 | mA |
| Emitter to base breakdown voltage | 3026 Cond. D | $I_E$ = -50 mAdc<br>$I_C$ = 0 | | $BV_{EBO}$ | -2 | --- | Vdc |
| Collector to emitter breakdown voltage | 3011 Cond. D | $I_E$ = -100 mA peak<br>$I_B$ = 0<br>Pulsed 2/ | | $BV_{CEO}$ (sus) | -100 | --- | Volts |
| Subgroup 3 | | | 5 | | | | |
| Floating potential | 3020 | $I_E$ = 0<br>Voltmeter input resistance = 10 megohms<br>$V_{CB}$ = -160V | | $V_{EBF}$ | --- | 0.5 | Volts |
| Forward-current transfer ratio | 3076 | $I_C$ = -1.0A<br>$V_{CE}$ = -2V | | $h_{FE}$ | 50 | --- | --- |
| | | $I_C$ = -10A peak<br>$V_{CE}$ = -2V<br>Pulsed 2/ | | $h_{FE}$ | 25 | 100 | --- |
| Collector to emitter saturation voltage | 3071 | $I_C$ = -1.0 Adc<br>$I_B$ = -100 mAdc | | $V_{CE}$ (sat) | --- | -0.15 | Vdc |
| Collector to emitter saturation voltage | 3071 | $I_C$ = -10 Adc<br>$I_B$ = -1.0 Adc | | $V_{CE}$ (sat) | --- | -0.30 | Vdc |



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Table I - Group A inspection (Cont'd).

| Examination or Test | Conditions | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | MIL-STD-750 Method | Specific Conditions | | | Min | Max | |
| Subgroup 3 (Cont'd) | | | | | | | |
| Collector to emitter saturation voltage | 3071 | $I_C$ = -20 Adc<br>$I_B$ = -2.0 Adc | | $V_{CE}$ (sat) | --- | -0.5 | Vdc |
| Base-emitter saturation voltage | 3066<br>Cond. A | $I_C$ = -1.0 Adc<br>$I_B$ = -100 mAdc | | $V_{BE}$ (sat) | --- | -0.6 | Vdc |
| Base-emitter saturation voltage | 3066<br>Cond. A | $I_C$ = -10 Adc<br>$I_B$ = -1 Adc | | $V_{BE}$ (sat) | --- | -0.75 | Vdc |
| Base-emitter saturation voltage | 3066<br>Cond. A | $I_C$ = -20 Adc<br>$I_B$ = -2 Adc | | $V_{BE}$ (sat) | --- | -1.0 | Vdc |
| Subgroup 4 | | | 10 | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3306 | $I_C$ = -1.0 Adc<br>$V_{CE}$ = -10 V<br>f = 5 mc | | $h_{fe}$ | 2 | --- | --- |
| Pulse response | 3251<br>Cond. A | Figure 5 | | $t_r$<br>$t_s$<br>$t_f$ | ---<br>---<br>--- | 4<br>6<br>2.5 | µsec<br>µsec<br>µsec |
| Subgroup 5 | | | 15 | | | | |
| Forward-current transfer ratio | 3076 | $I_C$ = -1.0 A<br>$V_{CB}$ = -2 V<br>$T_A$ = -40°C | | $h_{FE}$ | 35 | --- | --- |

1/ Pulse test: PW ≤ 1 msec, duty cycle ≤ 2 percent.
2/ Sweep test: 1/2 sinewave, 60 cps minimum.



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| 3005 | Prebulse test conditions: | --- | --- | --- | --- | Burnout by pulsing |
|---|---|---|---|---|---|---|

Table II - Group B inspection.

| Examination or Test | Conditions | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | MIL-STD-750 Method | Specific Conditions | | | Min | Max | |
| Subgroup 1 | | | 20 | | | | |
| Physical dimensions | 2066 | Figure 1 | | | | | |
| Subgroup 2 | | | 15 | | | | |
| Solderability | 2026 | | | | | | |
| Thermal shock (temperature cycling) | 1051 Cond. B | 5 cycles T = +110°C. | | | | | |
| Thermal shock (glass strain) | 1056 Cond. B | | | | | | |
| Moisture resistance | 1021 | Omit initial conditioning | | | | | |
| Endpoints: (See subgroup 4) | | | | | | | |
| Subgroup 3 | | | 15 | | | | |
| Shock | 2016 | 500 G (See 4.5.3) 5 blows each orientation $X_1$, $Y_1$, $Y_2$, $Z_1$ Pulse duration 1 msec | | | | | |
| Vibration fatigue | 2046 | Nonoperating | | | | | |
| Vibration, variable frequency | 2056 | 10 G | | | | | |
| Constant acceleration | 2006 | 2000 G each orientation $X_1$, $Y_1$, $Y_2$, $Z_1$ | | | | | |
| Endpoints: (See subgroup 4) | | | | | | | |
| Subgroup 4 | | | 15 | | | | |
| Salt atmosphere (corrosion) | 1041 | | | | | | |
| Endpoints: (subgroup 2, 3 and 4) | | | | | | | |
| Collector to base cutoff current | 3036 Cond. D | $V_{CB}$ = -160 Vdc $I_E$ = 0 | | $I_{CBO}$ | -- | -10 | mAdc |
| Forward-current transfer ratio | 3076 | $I_C$ = -1.0 A $V_{CE}$ = -2 V | | $h_{FE}$ | 50 | --- | --- |



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Table II - Group B inspection (Cont'd).

| Examination or Test | Conditions | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | MIL-STD-750 Method | Specific Conditions | | | Min | Max | |
| Subgroup 5 | | | | | | | |
| High temperature life (nonoperating) | 1031 | $T_{stg}$ = 100 $^{+0°}_{-3°}$ | λ=10% | | | | |
| Endpoints: (see subgroup 6) | | | | | | | |
| Subgroup 6 | | | | | | | |
| Steady state operation life | 1026 | $T_C$ = 90°C min.<br>$P_C$ = 20 W<br>$I_C$ = 1 Adc | λ=20% | | | | |
| Endpoints: (for subgroups 5 and 6) | | | | | | | |
| Collector to base cutoff current | 3036 Cond. D | $V_{CB}$ = -160 Vdc<br>$I_E$ = 0 | | $I_{CBO}$ | --- | -20 | mAdc |
| Forward-current transfer ratio | 3076 | $I_C$ = -1.0 A<br>$V_{CB}$ = -2 V | | $h_{FE}$ | 40 | --- | --- |



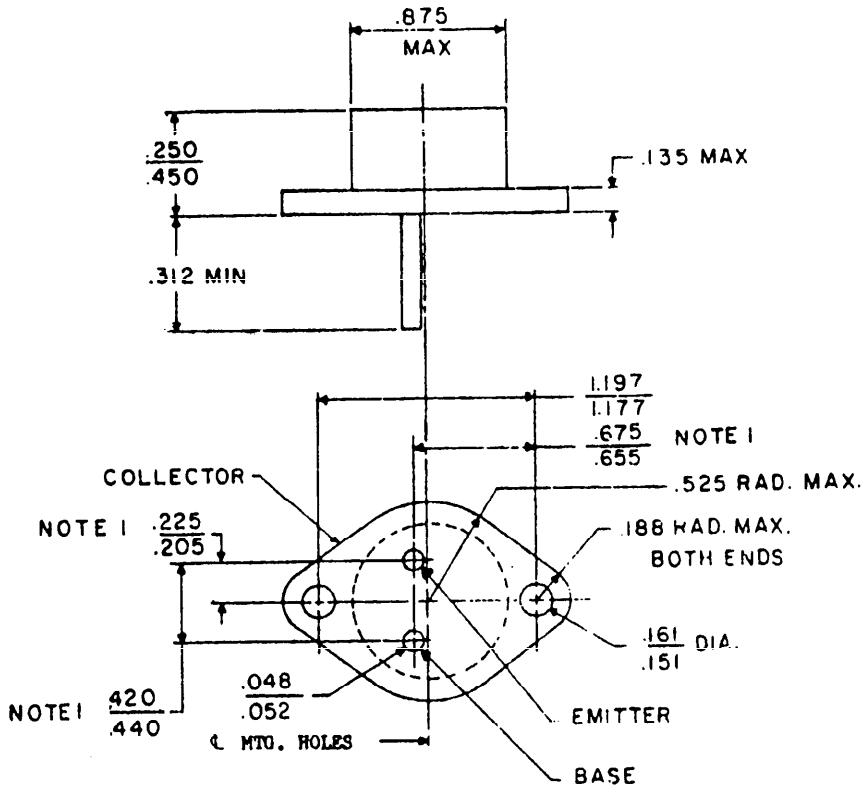
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Table III - Group C inspection.

| Examination or Test | Conditions | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | MIL-STD-750 Method | Specific Conditions | | | Min | Max | |
| Subgroup 1 | | | | | | | |
| Steady state operation life | 1026 | $T_C$ = 90°C min.<br>$P_C$ = 20 W<br>$I_C$ = 1 Adc | λ=10% | | | | |
| End Points: | | | | | | | |
| Collector to base cutoff current | 3036 Cond. D | $V_{CB}$ = -160 Vdc | | $I_{CBO}$ | --- | -20 | mAdc |
| Forward current transfer ratio | 3076 | $I_C$ = -1.0 A<br>$V_{CB}$ = -2V | | $h_{FE}$ | 40 | --- | --- |
| Subgroup 2 | | | 15 | | | | |
| Barometric pressure reduced | 1001 | 15 mmHg | | --- | --- | -10 | mAdc |
| Monitor collector to base cutoff current) | 3036 Cond. D | $V_{CB}$ = -160 | | | | | |
| Thermal resistance | 3151 | | | $\theta_{J-C}$ | --- | 1.0 | °C/W |
| Subgroup 3 | | | 15 | | | | |
| Safe Area Test 1 | --- | Test 1 (See 4.6.1, 6.2 and figure 2)<br>$V_{CE}$ = -50 V<br>$I_C$ = -20 A<br>PW = 50 μsec<br>f = 200 cps | | | | | |
| Safe Area Test 2 | --- | Test 2 (see 4.6.1, 6.2 and figure 2)<br>$V_{CE}$ = -150 V<br>$I_C$ = -30 mA<br>PW = 50 μsec<br>f = 200 cps. | | | | | |
| Safe Area Test 3 | --- | Test 3 (See 4.6.1, 6.2 and figure 2)<br>$V_{CE}$ = -50 V<br>$I_C$ = -150 mA<br>t = 5 minutes min. continuous<br>$T_C$ = 60°C max. | | | | | |
| Endpoints: (Same as group C, subgroup 1) | | | | | | | |



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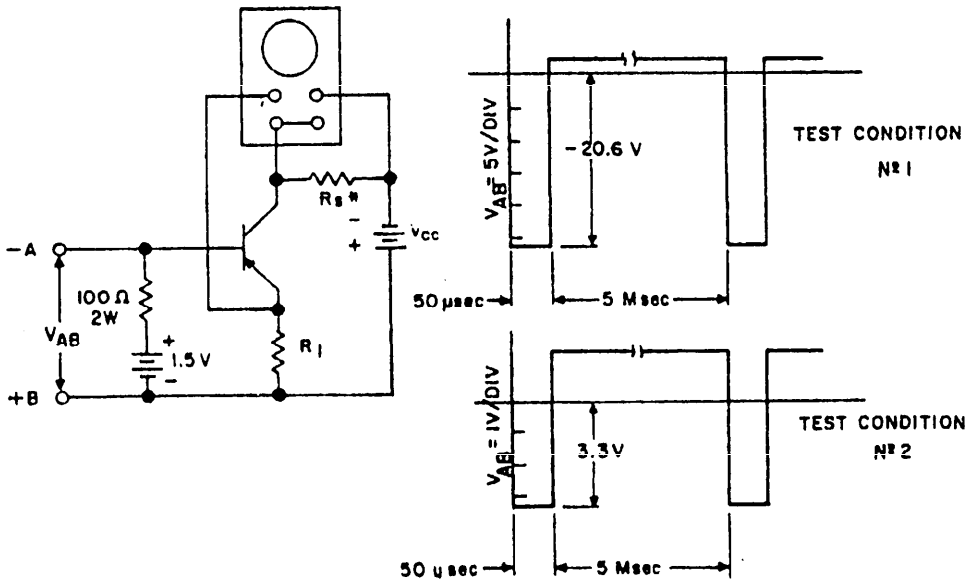

Notes:

1. These dimensions shall be measured at points 0.050 to 0.055 below seating plane. When gage is not used, measurement shall be made at base seat.
2. Unless otherwise specified, tolerances are ±0.005.
3. All dimensions are in inches.

Figure 1 - Outline dimensions of transistor type 2N2634.



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| TEST | $R_1$ | $V_{AB}$ | $V_{CC}$ |
|---|---|---|---|
| 1 | 1Ω 10W | (SEE TEST COND. Nº1) | 70 V |
| 2 | 100Ω 10W | (SEE TEST COND. Nº2) | 153V |
| 3 | 100Ω 5W | 16 V | 65V |

$V_{CC}$ AND $V_{AB}$ ARE REGULATED POWER SUPPLIES (± 1% BOTH LINE AND LOAD.)

* $R_S$ IS 0.1Ω NON INDUCTIVE 2 WATT. A CURRENT PROBE MAY BE USED IN LIEU OF THIS SENSE RESISTOR.

TEST Nº 3 A D.C. REGULATED SUPPLY OF APPROX. 16VOLTS IS CONNECTED TO POINTS A, B (MINUS TO POINT "A"). ADJUSTMENT WILL BE NECESSARY TO SET THE REQUIRED CONTINUOUS COLLECTOR CURRENT PER 4.6.1.

Figure 2 - Safe area test for transistor type 2N2834.



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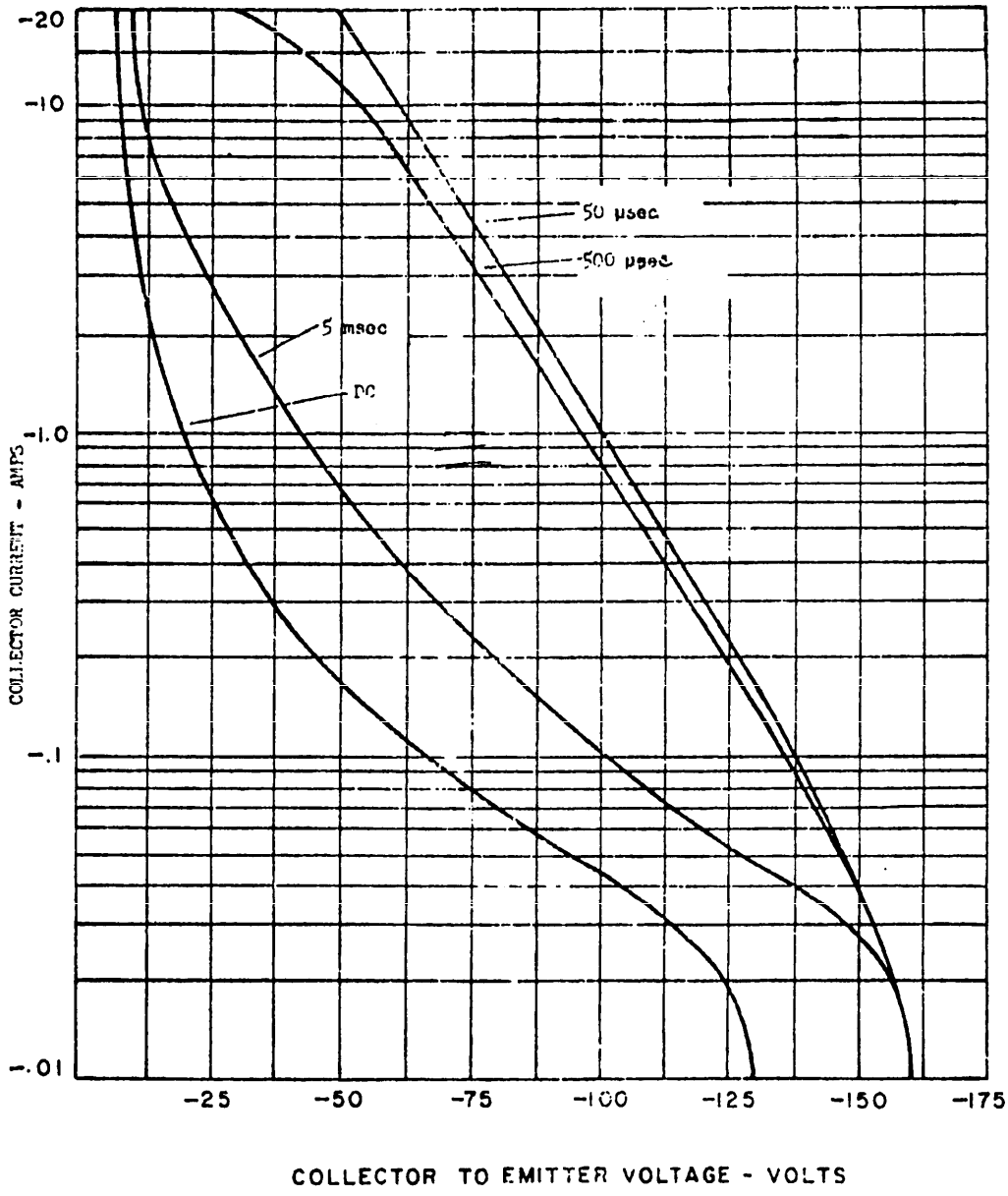



Figure 3 - Safe operating area of transistor type 2N2834.



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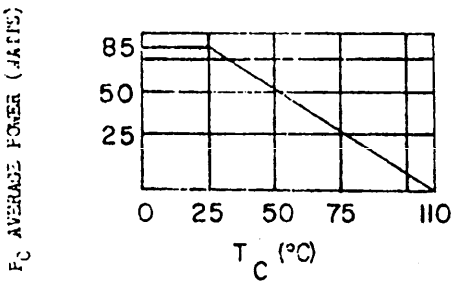



Figure 4 - Thermal derating curve.

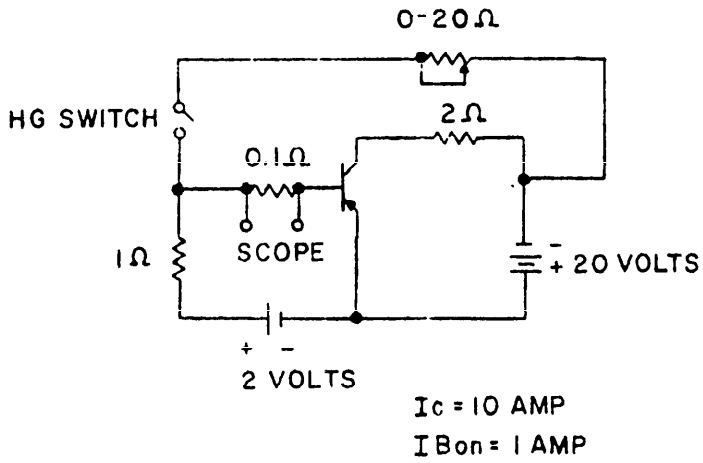



Figure 5 - Switching circuit.



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